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An Empirical Evaluation of Machine Learning Techniques for Chronic Kidney Disease Prophecy

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ABSTRACT Chronic Kidney Disease (CKD) implies that the human kidneys are harmed and unable to blood filter in the manner which they should. The disease is designated ''chronic'' in light of the fact that harm to human kidneys happen gradually over a significant time. This harm can make wastes to build up in your body. Many techniques and models have been developed to diagnos the CKD in early-stage. Among all techniques, Machine Learning (ML) techniques play a significant role in the early forecasting of different kinds ailments. ML techniques have been used for achieving analytical results which is one of the instruments utilize in medical analysis and prediction. In this paper, we employ experiential analysis of ML techniques for classifying the kidney patient dataset as CKD or NOTCKD. Seven ML techniques together with NBTree, J48, Support Vector Machine, Logistic Regression, Multi-layer Perceptron, Naïve Bayes, and Composite Hypercube on Iterated Random Projection (CHIRP) are utilized and assessed using distinctive evaluation measures such as mean absolute error (MAE), root mean squared error (RMSE), relative absolute error (RAE), root relative squared error (RRSE), recall, precision, F-measure and accuracy.The experimental outcomes accomplished of MAE are 0.0419 for NB, 0.035 for LR, 0.265 for MLP, 0.0229 for J48, 0.015 for SVM, 0.0158 for NBTree and 0.0025 for CHIRP. Moreover, experimental results using accuracy revealed 95.75% for NB, 96.50% for LR, 97.25% for MLP, 97.75% for J48, 98.25% for SVM, 98.75% for NBTree, and 99.75% for CHIRP. The overall outcomes show that CHIRP performs well in terms of diminishing error rates and improving accuracy.

INDEX TERMS CKD, ML techniques, CHIRP, MLP, NB, LR, J48, SVM, NBTree.

I. INTRODUCTION

Chronic Kidney Disease (CKD) is rising on regular basis and is widely considered to be a worldwide issue. This is a chronic issue related to improved mortality and bleakness, a high peril of a few other ailments including heart disease and human services cost [1]. CKD is also termed as chronic kidney failure or chronic renal failure, which is plentiful broad than individuals handle it regularly feelings un-watched and un-recognized till the disease is all around critical condition. Frequently, CKD is analyzed as a result of airing of individuals known to be in danger of issues, for instance, people with Diabetes, Heart Disease, High Blood Pressure, Obesity, Smoking, Hypertension, Alcohol Intake, Race/Ethnicity, Age and the individuals who have kinsfolks with CKD [2], [3]. All-inclusive, in excess of two million people get a kidney transplant or dialysis treatment to remain alive, so far this quantity can speak to just 10% of individuals who need treatment to live [4]. As indicated by the National Kidney Foundation,[1] in the United State there are twenty six million grown-ups who have CKD and millions others are at expanded hazard [5]. The formative phases of CKD are isolated into five phases. In the first phase, an individual can develop under customary kidney works and even

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experience minor mischief in kidney work. During second phase, an individual can encounter paltry to unobtrusive misfortune in kidney work. Phase three also misrepresents, with an individual encountering moderate to serious misfortune in kidney work. In fourth phase, an individual will encounter an extreme misfortune in kidney work. In Phase 5, an individual will encounter total kidney failure [5], [6]. It is also observed that the health issue because of CKD is likely to suffer to increment with the maturing populace and internationally ascend in Type 2 Diabetes [7]. Thus, it is expressively significant in the early controlling, overseeing and diagnosing of the ailment. It is imperative to find CKD disease precisely with a low error rate due to its active and undercover nature in the beginning periods, and patient heterogeneity.

Researchers are trying to use different Machine Learning (ML) techniques to overcome the limitations of early diagnosis of different kinds of diseases. Consequently, ML can provide a significant job in extracting important and concealed data from the colossal measure of the clinical and medical dataset. ML techniques are utilized in different fields to classify, cluster, predict and filter the data. Previously, different ML techniques are applied for the early prediction of disease, however still, there is uncertainty in the results of previously employed techniques [8].

In this paper, we present an empirical analysis of seven different ML techniques that are J48 Decision Tree, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Nevie Bayes Tree (NBTree), Logistic Regression (LR), and Navie Bayes (NB), and Composite Hypercube on Iterated Random Projection (CHIRP). The experimental outcomes of each technique is contrasted with each other. To evaluate the each technique, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Relative Squared Error (RAE), Root Relative Squared Error (RRSE), Precision, Recall, F-measure and Accuracy are used as evaluation measure. The overall outcomes of the study recommends that CHIRP outperforms to reduce the error rate and increase the accuracy for prediction of CKD. CHIRP is an intricate solution to overcome the limitations of other utilized techniques. This is a non-parametric outfit classifier that is not modified in each dataset, subsequently, it can achieve random information without their structure past data.

The major contributions of this research are as follow:

- We compare seven ML techniques (J48, SVM, MLP, NBT, LR, NB and CHIRP) for CKD prophery.
- We conduct a sereise of experiments on CKD dataset available on UCI repository.
- To provide insight into the experimental results, evaluation is carried out using MAE, RMSE, RAE, RRSE, Precision, Recall, F-measure and Accuracy.

The fundamental focal point of this emperical analysis is likewise proposing CHIRP for CKD prediction that has not been utilized for this issue in the prious studies.

Hereinafter; Section II presents the literature review, Section III contains the methodology, while experimental results and discussion are detailed in Section IV, and Section V discusses the overall conclusion.

II. LITERATURE REVIEW

CKD is a worldwide health issue that is rising on daily basis [9]. All around the world, about 10% of people are distress and early basis millions of people are dying of CKD [10]. There are a number of studies that accomplished to predict and diagnose CKD by utilizing ML techniques [11]. These techniques have been utilized for classification and prediction in the biomedical field [12]. A figure of momentous researches has prepared use of ML techniques that are illuminated here:

A. K-NEAREST NEIGHBOR

K-Nearest Neighbor (KNN) is employed by many researcher for classification problems. Hashi *et al.* [13] have used KNN for CKD on the dataset taken from Pima Indians data. Their results show that KNN produces accuracy of 76.96% with minimum error rate for CKD prediction. Bashir *et al.* [14] design a medical decision support framework using multi-layer classifiers for disease prediction. They also utilized KNN for early prediction of disease. Using KNN they achieved 57.41% accuracy on Statlong dataset and 58.42% accuracy on Cleveland dataset. Khan *et al.* [15] employed KNN for liver disease prediction on dataset taken from UCI ML repository. Their outcome of KNN shows 62.90% accuracy rate and 0.3718 for error rate.

B. DECISION TREE

Decision Tree (DT) is basically used for classification problems with ability to produce optimal results. Alasker *et al.* [12] employed DT for kidney disease prediction. They perform all the experiments on the dataset taken from UCI ML repository. Utilizing all the 24 features of dataset the achieved 98.4127% accuracy and 0.0159 of error rate, while utilizing 8 features they achieved 98.4127% accuracy and 0.0159 of error rate. Dar and Azmeen [16] uses this techniques for dengue fever prediction on real data taken from different hospitalized patients. Their outcome utilizing DT shows 76% accuracy to classify the data as positive or negative. Archarya *et al.* [17] utilized DT for fatty liver disease on ultrasound images. They have used one hundred ultrasound images in their research. The results achieved show that DT outperforms other techniques in terms of producing 99% accuracy.

C. ARTIFICIAL NEURAL NETWORK

Artificial Neural Network (ANN) is a very famous and mostly recommended technique utilize for classification problems. Baitharu and Pani [18] employed this techniques for healthcare support system using liver disease dataset. They have evaluated their system using error rate and accuracy measures. Their outcomes show the best accuracy of ANN which is 71.59% and lesser amount of error rate that is 0.3543. Vijayarani *et al.* [11] uses this ANN to kidney disease prediction on the dataset collected from different medical centers, labs and hospitals. According to their achieved results, ANN classification accuracy is 87.70%. Zhou *et al.* [19] employed artificial bee colony and particle swarm techniques for optimizing a neural network in prediction of heating and cooling loads of residential buildings. Using ANN on training data the error rate achieved is 2.5724 and on testing data it is 2.4275.

D. PROBABILISTIC NEURAL NETWORK

Probabilistic Neural Network (PNN) is a feed forward neural network containing three layers that are input layer, hidden layer and output layer. Archarya *et al.* [20] employed PNN to develop a decision support system for fatty liver disease prediction using ultrasound images. They perform all the experiments on the data collected form University of Malaya Medical Center, Malaysia. Their study recommended PNN as a best classifier with 98% accuracy. Rady and Anwar [1] employed PNN for prediction of kidney disease stages. In the conclusion they have suggested that PNN outperforms other employed techniques with 96.7% accuracy.

E. NAÏVE BAYES

Naïve Bayes (NB) is a probabilistic classification techniques extensively use for prediction of various types of diseases. Alasker *et al.* [12] worked out on NB for detecting kidney disease on the dataset taken from UCI ML repository that contains 24 attributes and 400 population. Among these population 150 are NOTCKD and the rest of 250 are in early phase of CKD. Utilizing all 24 attributes they achieved 99.3671% accuracy and 0.0057 of error rate. Ameer *et al.* [21] uses NB for author profiling for age and gender using different types of features. They have concluded their results for gender prediction on hotel reviews using Tn-gram (POS) features, that accuracy of NB is 60.6% and using Sn-gram (POS) features, accuracy of NB is 50.4%.

F. RANDOM FOREST

Random Forest (RF) is a cooperative learning method used for regression and classification problems. Khan *et al.* [15] use RF for diagnosing liver disease utilizing the dataset taken from UCI ML repository. They have concluded their results for RF with better accuracy of 72.17% and 0.3803 error rate. Carvajal *et al.* [22] employed RF to reveal the temporal pattern of dengue incidence using meteorological factors in metropolitan Manila, Philippines. They use two different datasets for experiments, one is Meteorological Factors and other is Lagged MF. On their experiments, RF error rate is 0.23 on Meteorological Factors dataset and 0.15 on Lagged MF dataset.

G. AdaBoost

AdaBoost (AB) is basically the arrangement of feeble techniques to develop a vigorous technique. Acharya *et al.* [20] use AB for decision support system for fatty liver disease using GIST descriptors extracted from ultrasound images. The experimental analysis using AB achieved 94% accuracy and 92% sensitivity. Acharya et al, also use AB for identification of fatty liver disease using random transform and discrete cosine transform features in ultrasound images. Yip *et al.* [23] utilize AB to develop a laboratory parameter-based non-alcoholic fatty liver disease in the general population.

H. SUPPORT VECTOR MACHINE

Support Vector Machine (SVM) is an influential learning technique that is grounded on current improvements in statistical learning theories. Pahareeya *et al.* [24] employed SVM for liver patient classification on data taken from UCI ML repository. On original dataset they have achieved 71.5026% accuracy while on over sampling they achieved 68.64% accuracy. Al-Anazi and Gates [25] employed SVM to classify lithofacies and model permeability in heterogeneous reservoirs. Utilizing SVM on log clustering based permeability training error rate is 1.1497, on core clustering based permeability training error rate is 0.6377, on log clustering permeability prediction error rate is 1.2885, and on core clustering based permeability prediction error rate is 1.5179.

III. METHODOLOGY

This paper is centered around the usage of ML techniques because of its procedures of value, distinguishing hidden patterns and providing novel information [26]. ML is additionally seen as a developing innovation that has made profound situated adjustments in the data biosphere [18]. Seven distinctive ML methods; NBTree, SVM, J48, MLP, LR, and NB, that are used in the early past along with CHIRP, utilized in this study to figure out which system will give the better classification outcomes in terms of accuracy and error rate, in order to analyze CKD in its beginning period. All these experiments are done through the methodology shown in Figure 1. In the initial step, data preprocessing is done as there is a part of missing values in the dataset taken form UCI ML repository. Missing values are dealt with utilizing, taking the mean of existing values and supplanting the missing fields with these mean values. After this progression, ML techniques are employed to the dataset, and afterward, the outcomes accomplished are examined utilizing accuracy and error rates.

A. NAÏVE BAYES

The Naïve Bayes (NB) classifier is essentially grounded on Bayes's hypothesis with autonomy suspicions among predictors [18]. NB model is exceptionally easy to fabricate and can be executed for colossal datasets. NB classifier frequently accomplishes well than increasingly refined classification strategies. The back likelihood, $P(A|B)$ is figured from $P(A)$, $P(B)$, and $P(B|A)$ as follows:

$$P(A|B) = \frac{p(B|A)p(A)}{p(B)} \tag{1}$$

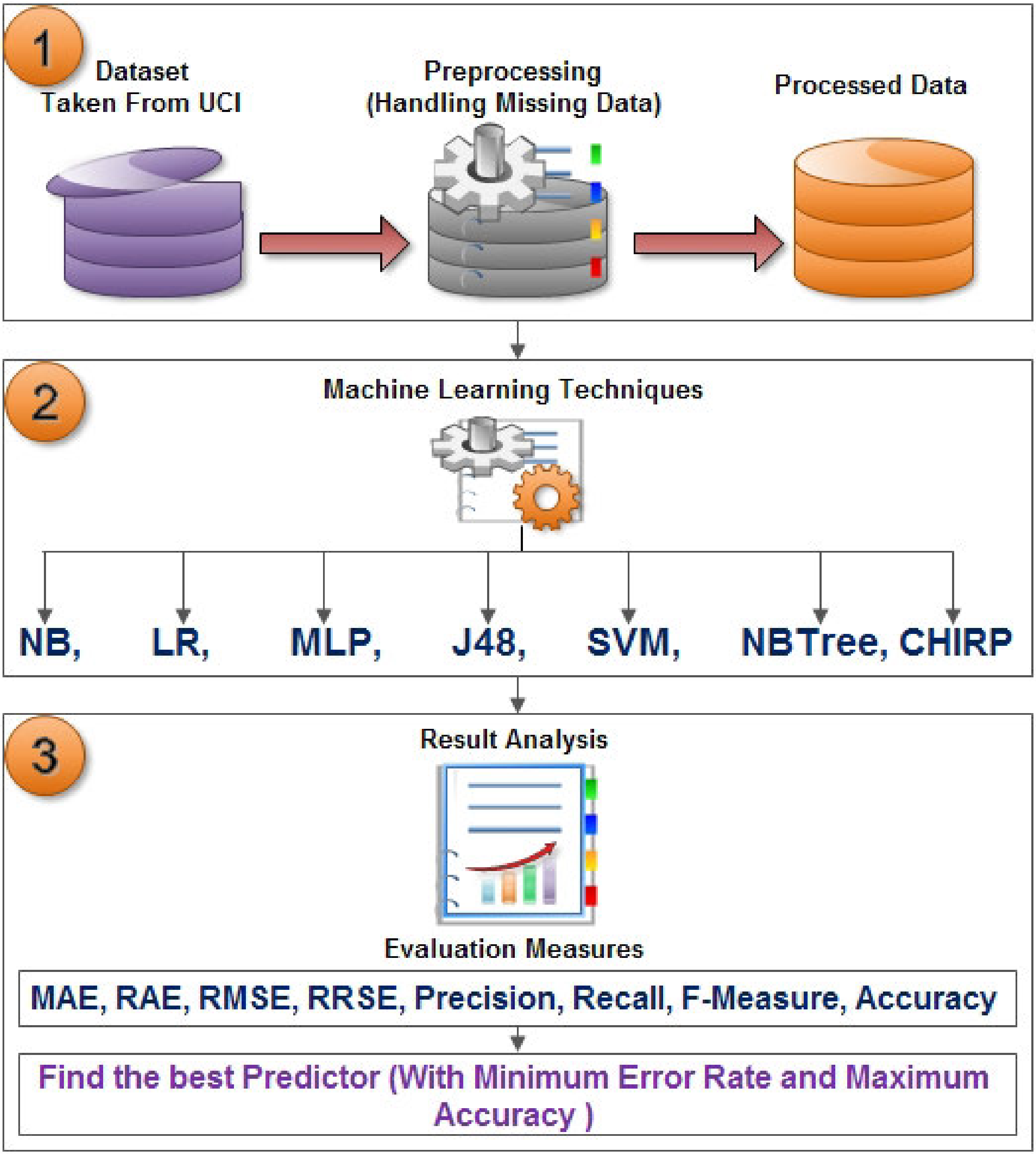

FIGURE 1. Methodology work-flow.

Using Bayesian probability terminology, the above equation can be written as:

$$Posterior = \frac{Prior * Likelihood}{Evidence} \tag{2}$$

here, *Posterior* is what parameters are likely after observing the data objects, *Prior* is uncertinity of data obejct prior, and *Likelihood* is the probability of falling under a specific category or class, while *Evidence* is the factor that is the same as for all possible hypothesis being considered.

The impact of the estimation of a forecaster (B) on an assumed class (A) is autonomous of the estimations of different predictors. This supposition is called class uncertain independence [27].

B. LOGISTIC REGRESSION

Logistic Regression (LR) quantifies the association between a continuous dependent variable and at least one independent variable, which are for the most part (yet not really) continuous, by utilizing likelihood scores as the predicted value of the

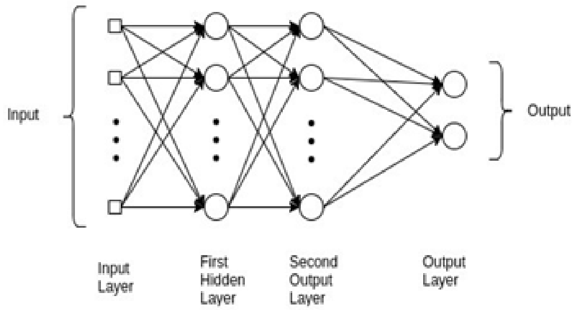

FIGURE 2. Example of MLP neural networks.

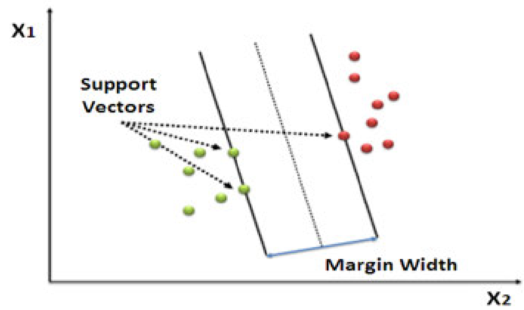

FIGURE 3. Example of SVM.

dependent variable [28]. Chances are the proportion between the probability of success over the probability of failure, that is, $pi/(1 - pi)$, though p is the probability of a model having a place with Class 0. At the point when $p > 0.5$, an example ought to be named Class 0. Else it ought to make a decision as Class 1. Since the calculated output probabilities dependent on the accompanying condition:

$$logit(p_{\mathrm{i}}) = log(\frac{p_{\mathrm{i}}}{1 - p_{\mathrm{i}}}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_{\mathrm{k}} x_{\mathrm{k}} \quad (3)$$

- the coefficients refer to each βi. Chances proportions are just the exponential of the weights we found previously. The coefficients are in certainty the weights that are applied to each attribute before including them together. In any case, the outcome is the probability that new occasion has a place with class yes (> 0.5 methods yes).

C. MULTI-LAYER PERCEPTRON

The Multi-Layer Perceptrons (MLPs) are considered utmost significant classes of neural networks, comprising of an input layer, minimum single hidden layer, and the output layer, as revealed in Figure 2. MLPs have been employed viably to tackle troublesome and different issues, by means of working out them in a supervised way using a well-known technique i.e., the back-proliferation algorithm [29]. This technique depends on the error rectification learning instructions. All things considered; it might be seen as speculation of a versatile filtering algorithm.

D. J48 DECISION TREE

J48 is an advance rendition of C4.5. The methodology of this technique is to utilize the divide and conquer strategy. It utilizes a pruning strategy to develop the tree. It is a typical technique that is used in information gain or entropy measures. Hence, it resembles tree structure with the root node, moderate and leaf nodes. Nodes hold the decision and gain the outcome [30].

E. SUPPORT VECTOR MACHINE

The SVM is a influential learning algorithm dependent on ongoing developments in the arithmetical learning hypothesis [31]. SVM is a discriminative classifier properly characterized by an isolating hyper-plane. So on, given supervised training data, the algorithm results an ideal hyperplane that arranges novel models. In dual dimensional space, this hyperplane is a line separating a plane in dual sections wherein individual class lays on either side [32] as appeared in Figure 3.

F. NBTree

This is a hybrid algorithm of NB and DT. The algorithm is like the traditional recursive parceling plans, then again, actually, the leaf nodes made are NB categorizers rather than nodes anticipating a solitary class [33].

G. COMPOSITE HYPERCUBE ON ITERATED RANDOM PROJECTION

Composite Hypercube on Iterated Random Projection (CHIRP), is an iterative component of three phases; anticipating, covering, and binning, which expected to a settlement with the scourge of dimensionality, computational unpredictability, and nonlinear recognizability [34]. This technique is not the hybridization of different models, additionally not the improvement or adjustment of winning models; it uses new packaging methods. The precision of CHIRP on commonly used objective datasets outperforms the accuracy of contenders. The CHIRP utilizes the computationally compelling ways to deal with assembling 2D forecasts and sets of quadrangular areas on those forecasts that involve assessments from an individual group of data. CHIRP classifies these groups of predictions and sections them into an end slant for tallying new data opinion [35].

IV. RESULTS AND DISCUSSION

The accompanying analysis is performed using different ML methods on the dataset taken from the UCI ML repository, which is already have been used in the previous researches [36]. The exploratory consequences of the employed methods are completed dependent on the performance measure of the classification error rate and accuracies. Model testing and approval are achieved by the N-fold cross-validation procedure.

Missing values in the dataset are supplanted by medians in the experiment. The dataset utilized in this study comprised of 400 instances in which 250 are CKD and 150 are NOTCKD instances, with overall 25 attributes that are

TABLE 1. List of CKD dataset attributes and description.

S. No.	Attribute	Description	S. No.	Attribute	Description
1	age	Age in years	14	pot	Potassium (Numerical)
2	bp	Blood Pressure (Numerical)	15	hemo	Hemoglobin (Numerical)
3	sg	Specific Gravity (Nominal)	16	pcv	Packed Cell Volume (Numerical)
4	al	Albumin (Nominal)	17	wc	White Blood Cells Count (Numerical)
5	su	Sugar (Nominal)	18	rc	Red Blood Cells Count (Numerical)
6	rbc	Red Blood Cells (Nominal)	19	htn	Hypertension (Nominal)
7	pc	Pus Cell (Nominal)	20	dm	Diabetes Mellitus (Nominal)
8	pcc	Pus Cell Clumps (Nominal)	21	cad	Coronary Artery Disease (Nominal)
9	ba	Bacteria (Nominal)	22	appet	Appetite (Nominal)
10	bgr	Blood Glucose Random (Numerical)	23	pe	Pedal Edema (Nominal)
11	bu	Blood Urea (Numerical)	24	ane	Anemia (Nominal)
12	sc	Serum Creatinine (Numerical)	25	class	Class (Nominal)
13	sod	Sodium (Numerical)			

TABLE 2. Attributes measurements and values range.

S. No.	Attribute	Measurement and Values Range	S. No.	Attribute	Measurement and Values Range
1	age	Age in Years	14	pot	mEq/L
2	bp	bp in mm/Hg	15	hemo	gms
3	sg	1.005,1.010,1.015,1.020,1.025	16	pcv	numerical values
4	al	0,1,2,3,4,5	17	wc	cells/cumm
5	su	0,1,2,3,4,5	18	rc	millions/cmm
6	rbc	normal, abnormal	19	htn	yes, no
7	pc	normal, abnormal	20	dm	yes, no
8	pcc	present, notpresent	21	cad	yes, no
9	ba	present, notpresent	22	appet	good, poor
10	bgr	mgs/dl	23	pe	yes, no
11	bu	mgs/dl	24	ane	yes, no
12	sc	mgs/dl	25	class	ckd, notckd
13	sod	mEq/L			

highlighted in Table 1 along with their descriptions. Among these 25 attributes, 11 attributes are numeric and the remainder of 14 attributes are nominal. The rest of the information regarding attribute values with the given range is mentioned in Table 2.

A. ASSESSMENT CRITERIA

MAE [37], [38], RMSE [39], [40], RAE percentage [41], and RRSE percentage [41] are utilized to assess the error rate of classification technique and Precision [42], Recall [43], F-Measure [16], and Accuracy [42], [44] percentage are used to measure the accuracy performance of employed techniques. All these assessments can be performed using the given conditions:

1) MEAN ABSOLUTE ERROR

Is the average of all absolute errors that can be calculated as:

$$MAE = \frac{1}{2}\sum_{j=1}^{n}|y_i - y| \tag{4}$$

where n is the number of errors, $|y_i - y|$ is the absolute error.

2) ROOT MEAN SQUARED ERROR

Is a quadratic scoring rule that also measures the average magnitude of the error.

$$RMSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_i - 1)^2} \tag{5}$$

3) RELATIVE ABSOLUTE ERROR

Is likewise comparative with a simple predictor, which is only the average of the real values. For this situation, however, the error is only the total absolute error rather than the overall squared error.

$$RAE = \frac{\sum_{j=1}^{n}|P_ij - T_j|}{\sum_{j=1}^{n}|T_j - \bar{T}|} \tag{6}$$

where $P_(ij)$ is the value forecast by the specific model I for record j (out of n records), Tj is the goal value for record ji and $\bar{T}$ is as follow;

$\bar{T} = \frac{1}{2}\sum_{j=1}^{n} T_j$, for a perfect fit, the numerator is equal to 0 and $Ei = 0$.

4) ROOT RELATIVE SQUARED ERROR

Is just the simple average of the real values. Subsequently, the relative squared error revenues all out the squared error and standardizes it by separating the entire squared error of the modest predictor. By taking the square root of the relative squared error individual diminutions the error to same dimensions as the quantity being forecasted.

$$RRSE = \sqrt{\frac{\sum_{j=1}^{n}(P_ij - T_j)^2}{\sum_{j=1}^{n}(T_j - \bar{T})^2}} \tag{7}$$

5) ACCURACY

Is the most instinctive performance measure and it is just a ratio of correctly classified or predicted perceptions to the overall perceptions.

$$Accuracy = TP + \frac{TN}{TP} + FP + FN + TN \tag{8}$$

where TP is the number of true-positive classifications, FN is the number of false-negative classifications, TN is the number of true-negative classifications, and FP is the number of false-positive classifications.

6) PRECISION

Is the ratio of effectively anticipated positive observations to all-out anticipated positive observations.

$$Precision = \frac{TP}{TP} + FP \tag{9}$$

7) RECALL (SENSITIVITY)

Recall is the ratio of accurately anticipated positive observations to all observations in genuine class – yes.

$$Recal = \frac{TP}{TP} + FN \tag{10}$$

8) F-MEASURE

Is the weighted average of Precision and Recall. Along these lines, this score considers both false negative and false positives. Instinctively it is not as straightforward as precision, however, F1-Measure is normally more helpful than accuracy, particularly in the event that you have a lopsided class distribution.

$$F - Measure = 2\frac{(Recall \times Precision)}{(Recal + Precision)} \tag{11}$$

The 10-fold cross-validation test is applied to every classifier to assess its performance. The 10-fold cross-validation is a procedure that divides the dataset into ten subsets of equivalent sizes; one subset is utilized for testing while the others are utilized for training. This is done until every subset has been utilized for testing. 10-fold cross-validation is the standard strategy for assessment [29].

B. EXPERIMENTAL RESULTS

Table 3 shows the results of all correctly and incorrectly classified instances by the techniques employed. The rows of the table show individual outcomes of each technique, while the column shows the overall outcomes of correctly and incorrectly instances using each technique. These outcomes show that CHIRP outperforms other techniques while correctly classifying 399 instances among 400.

TABLE 3. Analysis of correctly and incorrectly classified instances using different classifiers.

Technique	Correctly Classified	Incorrectly Classified
CHIRP	399	1
NBTree	395	5
SVM	393	7
J48	391	9
MLP	389	11
LR	386	14
NB	383	17

TABLE 4. Error rate assessment for all classifiers.

Technique	MAE	RAE %	RMSE	RRSE %
CHIRP	0.0025	0.5331	0.05	10.3279
NBTree	0.0158	3.3643	0.1011	20.8838
SVM	0.0175	3.732	0.1323	27.3252
J48	0.0229	4.889	0.1404	29.0057
MLP	0.265	5.6597	0.1406	29.0483
LR	0.035	7.4652	0.1871	38.6435
NB	0.0419	8.9383	0.1905	39.359

FIGURE 4. MAE and RAE assessment for all classifiers.

The error rate assessment for all classifiers is presented in Table 4. First column represents the name of used techniques while experimental results for error rates are listed from second to last columns. For all absolute errors (MAE and RAE%), the experimental results show the best performance of CHIRP in term of reducing the error rate.

Similarly, for squared errors (RMSE, RRSE%), experimental results reveal the better performance of CHIRP as compared to other employed techniques. Figure 4 shows the experimental results for all classifiers using MAE and RAE, while Figure 5 shows results for RMS and RRSE.

Table 5 shows the overall terminology of the confusion matrix. Table 6 shows the confusion matrix for all the assessments evaluated during experiments.

There are two types of classes in which prediction is possible, i.e., Class 1 (Positive) and Class 2 (Negative). If we forecast the occurrence of a disease, for example, Class 1 revenue

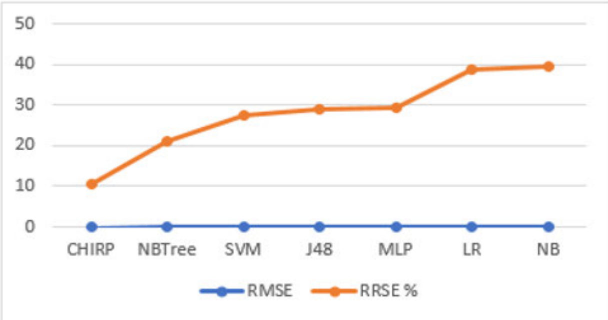

FIGURE 5. RMSE and RRSE assessment for all classifiers.

TABLE 5. Confusion matrix terminology.

	Class/Positive 1	Class/Negative 0
Class 1 or Positive (1)	TP	FP
Class 2 or Negative (0)	FN	TN

TABLE 6. Confusion matrix for all classifiers.

Technique	CKD	NOTCKD	Class
CHIRP	249	1	CKD
	0	150	NOTCKD
NBTree	246	4	CKD
	1	149	NOTCKD
SVM	243	7	CKD
	0	150	NOTCKD
J48	247	3	CKD
	6	144	NOTCKD
MLP	240	10	CKD
	1	149	NOTCKD
LR	238	12	CKD
	2	148	NOTCKD
NB	233	17	CKD
	0	150	NOTCKD

TABLE 7. Assessment of overall Precision, Recall, F-Measure and accuracy.

Technique	Precision	Recall	F-Measure	Accuracy %
CHIRP	0.998	0.998	0.998	99.75
NBTree	0.988	0.988	0.988	98.75
SVM	0.983	0.983	0.983	98.25
J48	0.978	0.978	0.977	97.75
MLP	0.974	0.973	0.973	97.25
LR	0.967	0.965	0.965	96.50
NB	0.962	0.958	0.958	95.75

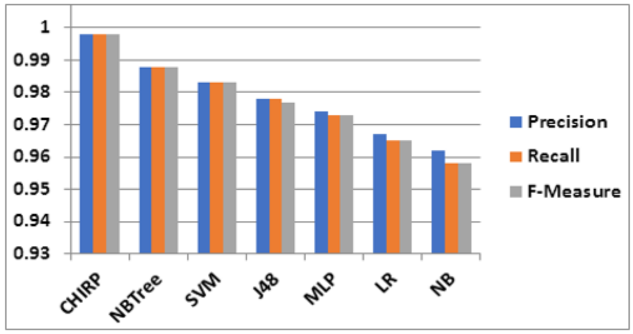

FIGURE 6. Analysis of Precision, Recall and F-Measure.

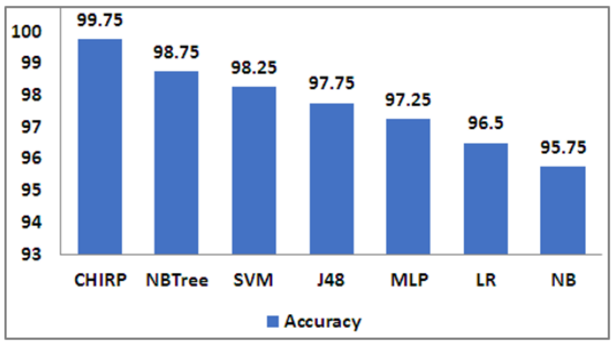

FIGURE 7. Accuracy evaluation of all classifiers.

that the person have the disease, and Class 2 revenue that the person does not have the disease. Here, TP is the case in which the patient as positive (they have the disease), FP is also the cases of positive, but they do not actually have the disease, and it is also called Type 1 error. FN is the negative cases, but they actually do have the disease, and it is also known as Type 2 error. TN is a negative case, which shows that they do not have the disease.

Based on this confusion matrix, recall, F-Measure, precision, and accuracy are determined as shown in Table 7, where the rows show individual assessment of every classifier while the columns show results of every assessment measure.

C. DISCUSSION

As mention in the analysis that CHIRP outperforms in contrast with other employed classifiers to reduce the error rate and increase the accuracy. Now, a question can ascend here that why CHIRP results are fruitful than other classifiers. Just as, there can be a few factors that may influence the validity of our approach. Both of these points are addressed in the subsequent sections.

1) WHY CHIRP RESULTS ARE FRUITFUL THAN OTHER CLASSIFIERS?

Supervised classifiers appearance some thoughtful issues, Fisrt one is the scourge of dimentionality: the closest relatives in multi-dimensional places separate more rapidly with measurement. A subsequent issue is a computational intricacy: polynomial-time techniques are not good in multi-dimensional places. A third issue is a separability: in some cases, a vector space (e.g., a 2D circle encompassed by a ring) there happens no equivalency that permits partition of sections utilizing hyper-planes in the codomain [34]. There are various ways to deal with defeating these issues, for example, hybridization of different classifiers [45] or decrease of dimensionality using main components, K-mean clustering [46], or arbitrary projections.

CHIRP is an integrated solution that tends to these issues. This technique is a non-parametric ensemble classifier technique that does not necessitate to be modified in individual dataset; so, it can manage various data deprived of their structure earlier information [34]. This classifier depends on a pictorial analytics framework [47] utilizing known as Composite Hypercube Description Regions (CHDRs) that can be utilized to characterize local and large scale structures [34], [48]. This classifier is intended to address the scourge of dimensionality and exponential multifaceted nature by utilizing binning, covering and projectionin a consecutive structure. For points that classes are labled in a multi-dimentional space, CHIRP utilizes arithmetically-efficient approaches to build 2D forecasts and groups of quadrilateral areas on individuals forecasts that comprise facts after just single class [34]. CHIRP arranges these groups of forecasts and areas addicted to a conclusion incline for counting innovative data points. It is average forecast accuracy over a extensive scope of datasets that surpasses that of modest classifiers, particularly ones that have complex constraints likewise pruning criteria, bandwidths, kernel types, and so forth [34], that is why this study proposes the CHIRP for the early and accurate prediction of CKD.

2) THREATS TO VALIDITY

In this subsection, we discuss the influences that may distress the validity of our approach.

Internal Validity: Our analysis in this paper is based on different well-known assessment criteria that are employed previously in different studies. Some of these criteria are to evaluate the error rate and some to evaluate the accuracy. Thus, the treat can be that replacement of new assessment criteria instead of employed criteria can reduce the accuracy. Moreover, the techniques employed in this research can be replaced with some latest techniques are can be hybridized with each other that can produce better results than the proposed technique.

External Validity: We conducted experiments on the dataset taken from the UCI ML repository containing 400 instances. A threat to validity may ascend because of, if we apply the proposed techniques in the more real data collected from the different hospitals using surveys etc. or the population size may rise in the dataset in the updated version, that may affect the results while increasing the error rates. As well, the proposed technique may not be able to produce better prediction in results using some other disease datasets. Therefore, this study focused on the CKD dataset to assess the performance of the proposed technique as compare to other utilized techniques.

Construct Validity: In this study, different ML techniques are compared with CHIRP on the base of multiple assessment measures. The selection of CHIRP is on the basis of its advanced features over the other classifiers that have been utilized by the researchers in the early past. However, the threat can be that if we apply some other new techniques, then it can be the possibility that they can beat CHIRP. Moreover, if we split the dataset into training and testing, as well as increasing or decreasing the number of fold validation for the experiments can also cause to reduce the error rate. It also can be possible that utilizing the latest assessment criteria produces better results that can beat the current achieved results.

V. CONCLUSION

Researches have reported that CKD results into the tens of thousands human death toll. It is therefore of great interest to find out ways to diagnos the CKD in early-stage. A number of ML algorithms have been employed to automatically classify CKD. Till now, no work has been done to compare the reported ML algorithms. In this paper, we mainly focus on the empirical comparisons of seven ML algorithm. For this purpose, we select NB, LR, MLP, J48, SVM, NBTree, and CHIRP. The CKD prophecy results of experiments show better performance for CHIRP on an average using different evaluation metrics.

Finally, as observed from Table 4 and Table 7, when comparing the overall error metrics, MAE, RAE%, RMSE, and RRSE% CHIRP performance is 0.0025, 0.5331, 0.05, and 10.3279 respectively which is the minimal error rate as compared to other employed techniques. Moreover, comparing overall accuracy metrics, precision, recall, F-measure CHIRP performance is 0.998 for all these three metrics and 99.75% for accuracy that show the better performance of CHIRP as contrasted with entire utilized techniques.

These results clearly highlight that NB, LR, MLP, J48, SVM, and NBTree may not appropriately handle the CKD prophecy and thus experimental results proved CHIRP as a promising technique for CKD prophecy.

SIGNIFICANCE STATEMENT

In this paper, we utilized seven ML techniques on the dataset that is taken from the UCI ML repository comprising 400 Instances. The outcomes of mentioned techniques have been contrasted with characterizing the most precise technique that brings about classifying the CKD and NOTCKD patients with high accuracy and less error rate. This study prescribed the CHIRP is the best technique that can be utilized by practitioners so as to eradicate diagnostic and treatment errors.

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